

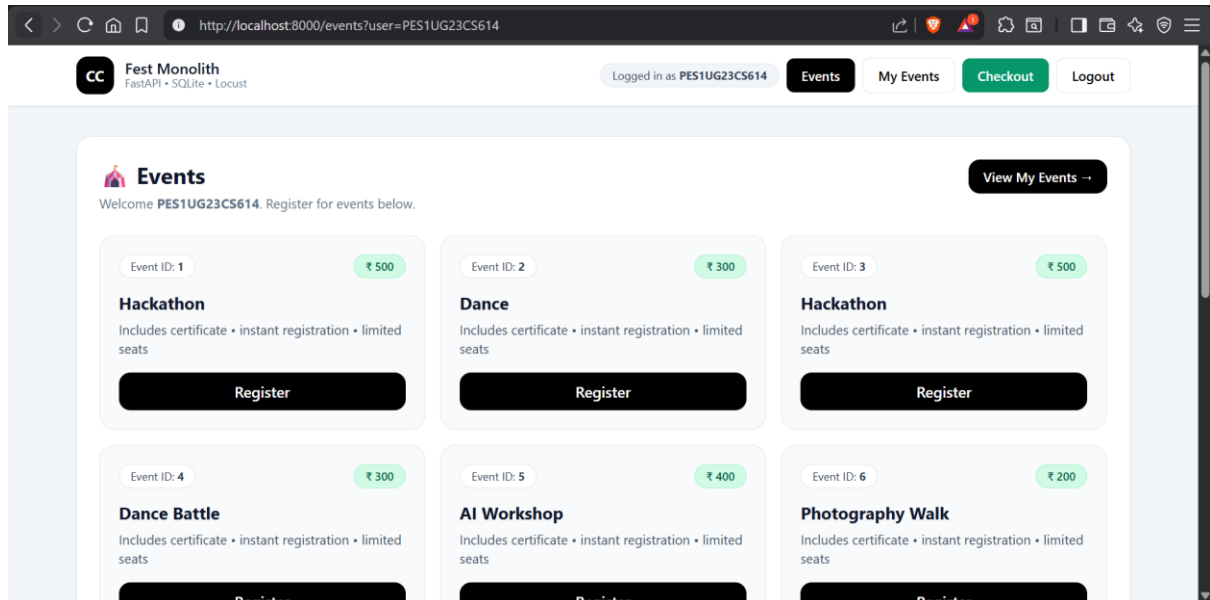
Name: Suprateek Yawagal

SRN: PES1UG23CS614

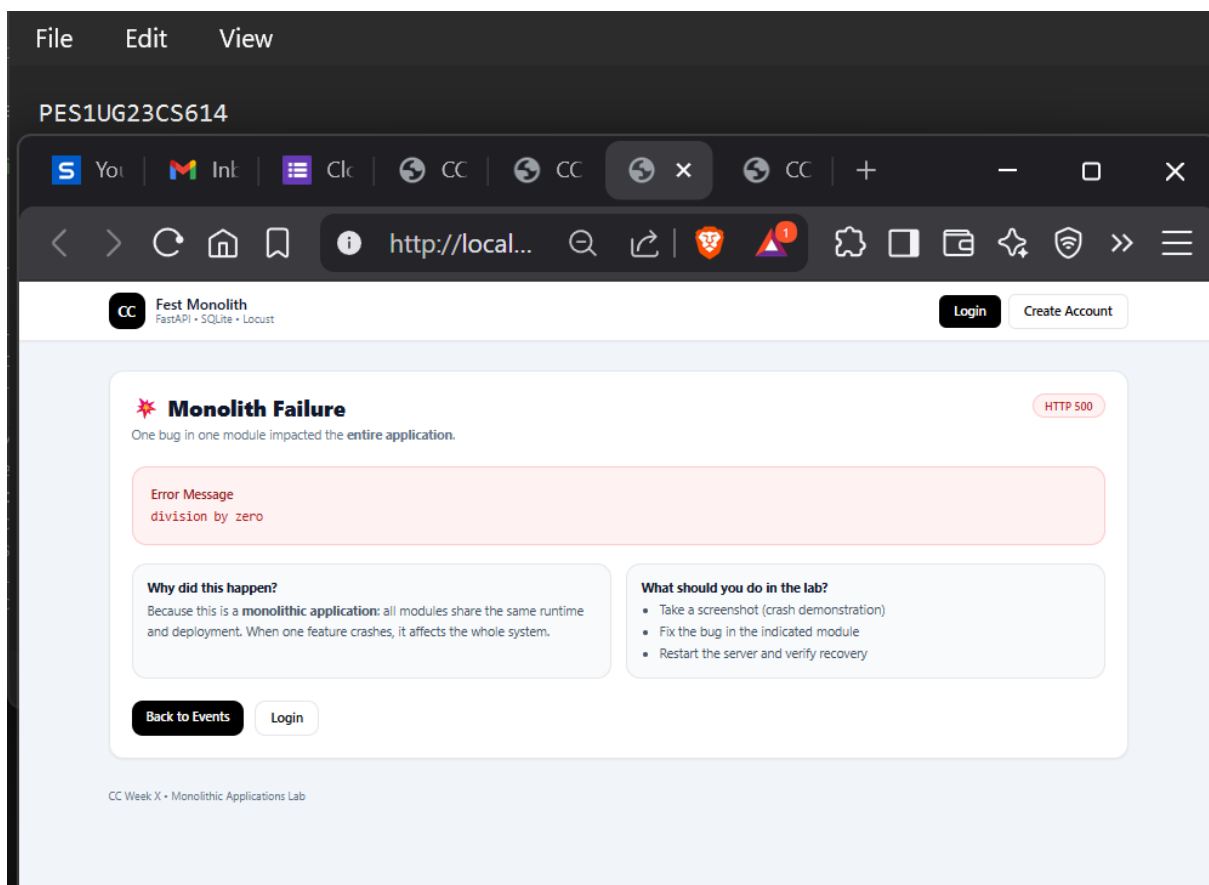
Sec: J

Github Repo : <https://github.com/bugsNburgers/CC-LAB-2-Monolithic>

Screenshot 1:

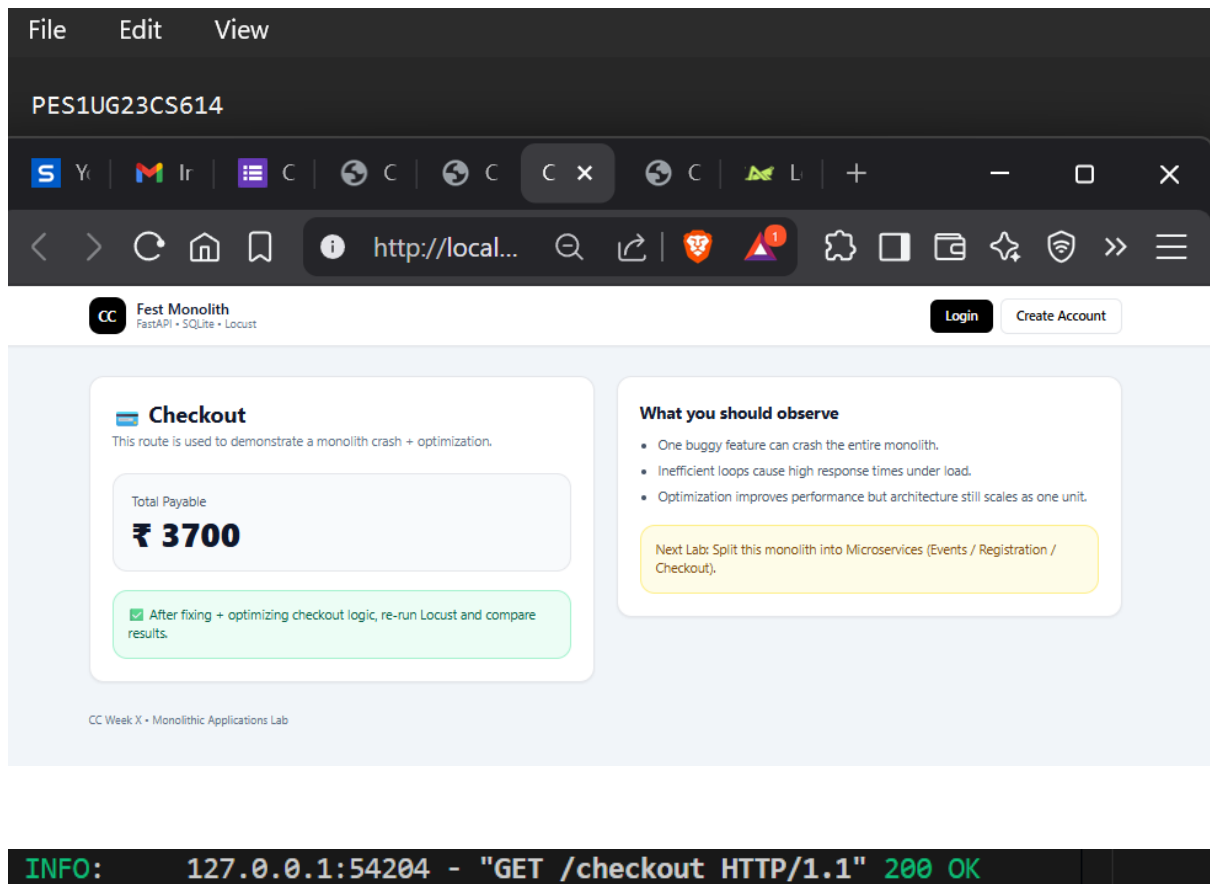


Screenshot 2:

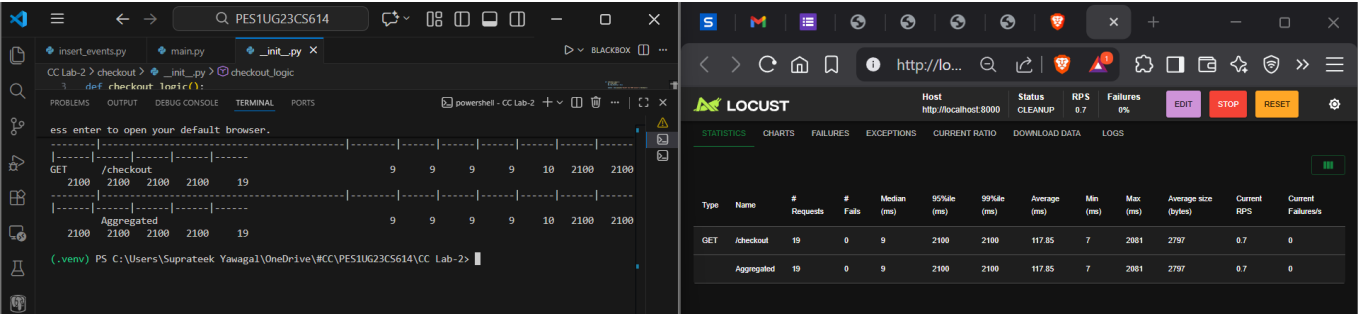


```
INFO:      127.0.0.1:64163 - "GET /checkout HTTP/1.1" 500 Internal
          Server Error
ERROR:     Exception in ASGI application
```

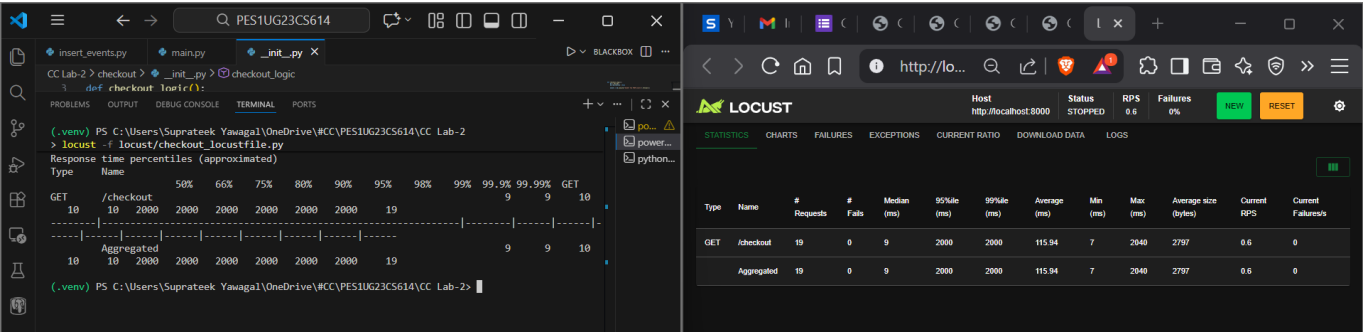
Screenshot 3:



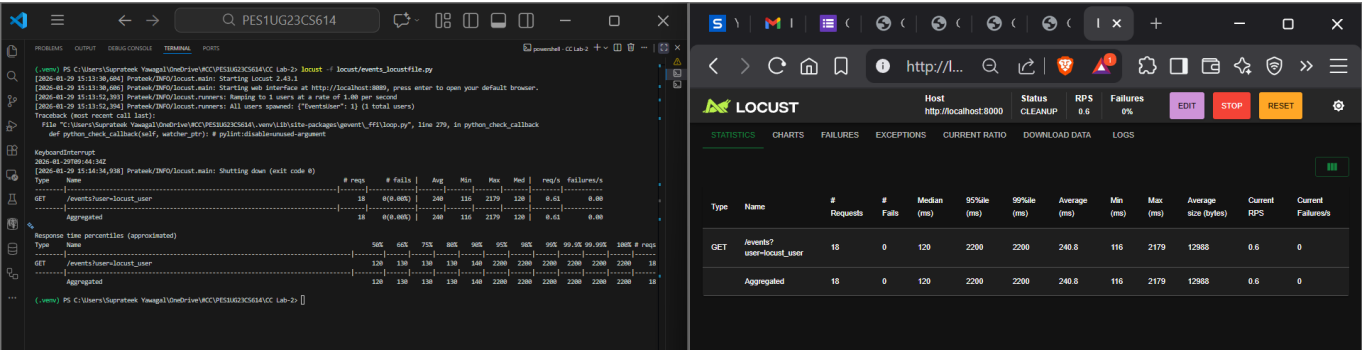
Screenshot 4:



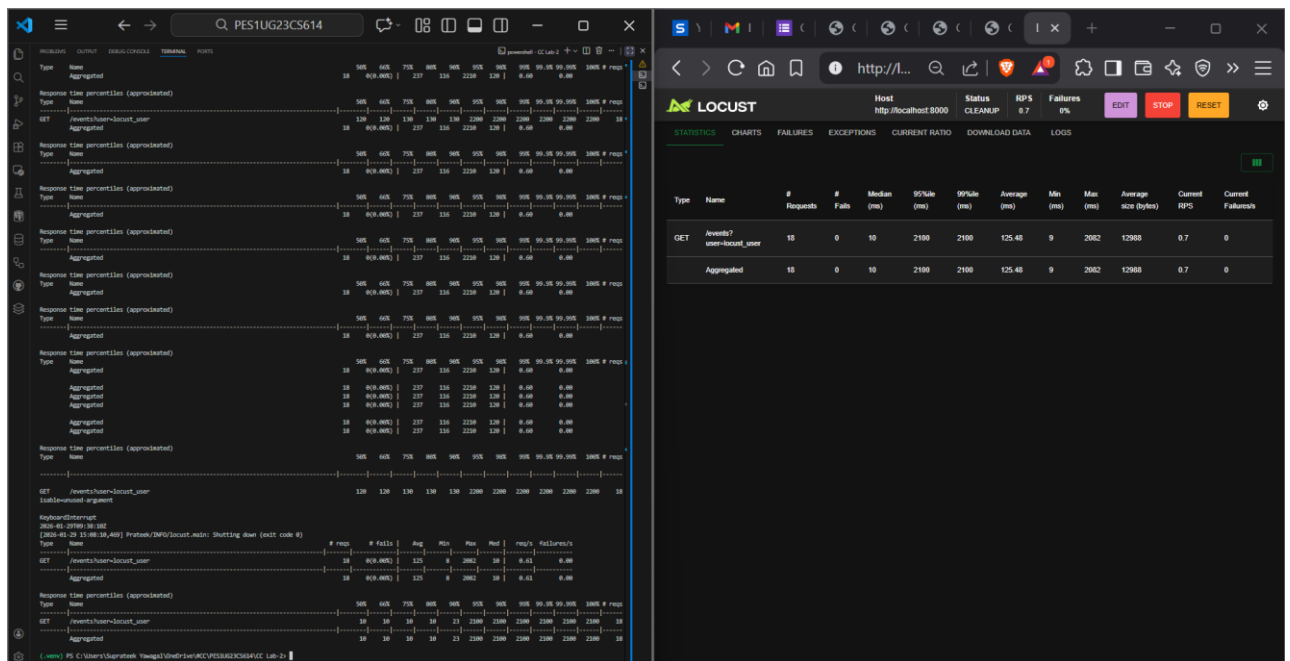
Screenshot 5:



Screenshot 6:



Screenshot 7:



Q: What was the bottleneck?

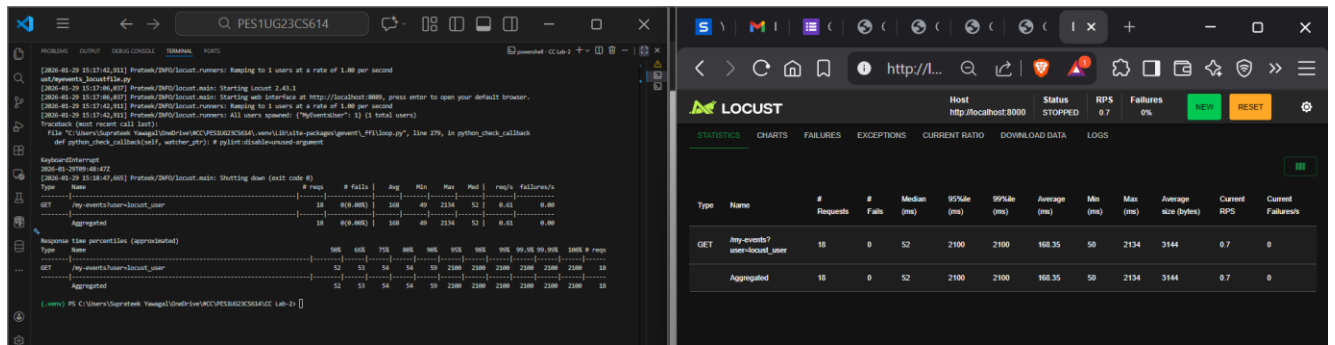
A: A wasteful computational loop iterating 3 million times, performing unnecessary modulo operations (`waste += i % 3`).

Q: What change did you make?

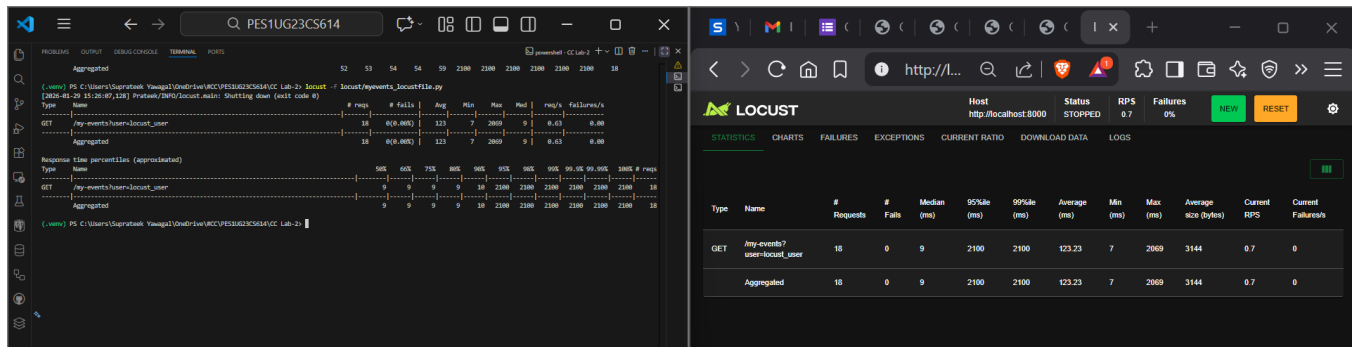
A: Removed the entire wasteful loop (`waste = 0` and `for i in range(3000000): waste += i % 3`).

Q: Why did the performance improve?

A: The route was spending significant CPU time on meaningless calculations that didn't contribute to the response. Removing this eliminated unnecessary CPU cycles, allowing the server to process requests much faster and handle more concurrent users.



Screenshot 9:



Q: What was the bottleneck?

A: A wasteful computational loop iterating 1.5 million times, performing unnecessary increments (dummy += 1).

Q: What change did you make?

A: Removed the entire wasteful loop (dummy = 0 and for _ in range(1500000): dummy += 1).

Q: Why did the performance improve?

A: The server was wasting CPU time on useless iterations. Removing this overhead reduced response time dramatically, increased throughput (requests/second), and allowed the application to scale better under load.